

CONSTRUCTION ESTIMATING CASE STUDY

Construction of a Two-Storey Residential Building with Swimming Pool

Samar, Philippines



Architectural perspective render, as issued in the drawing set (Sheet A.01)

Prepared by
Ma. Jessa P. Dollado, RCE, Master Plumber
Remote Construction Estimator · Quantity Surveyor · Technical Reviewer
Bluebeam · Excel · BOQ · Cost Estimation · Technical Review

Project Snapshot	
Type	Residential Construction, Philippines
Site	Coastal, flood-prone, Seismic Zone 4, typhoon corridor
Turnaround	48 hours — complete estimate package
Workflow	Bluebeam + Excel + Claude-assisted technical review

Executive Summary

Item	Detail
Project	Construction of a Two-Storey Residential Building with Swimming Pool
Client Goal	Sign-and-sealed, permit-ready structural and plumbing review; complete quantity takeoff and detailed cost estimate
My Role	Construction Estimator · Technical Reviewer · Quantity Surveyor · Documentation Lead
Duration	48-hour initial turnaround for the estimate package
Total Project Cost	₱7,586,231.52 (inclusive of OCM, profit, and VAT)
Contract Duration Assumed	120 calendar days
Building Footprint	≈ 138 sqm total floor area over 2 storeys, plus attached pool

Deliverables

- ✓ Drawing Review
- ✓ Technical (Structural/Geotechnical) Review
- ✓ Bluebeam Markups
- ✓ Quantity Takeoff
- ✓ Bill of Quantities (BOQ)
- ✓ Detailed Cost Estimate
- ✓ DUPA (Detailed Unit Price Analysis)
- ✓ RFI Log
- ✓ Drawing Revisions
- ✓ Final Estimate Package

Project Background

Client & Project

The client is a private homeowner constructing a two-storey residential building with an attached swimming pool in Barangay 3 Poblacion, San Julian, Eastern Samar — a coastal poblacion on the Pacific-facing seaboard of Samar Island. The client engaged me for two linked scopes: sign-and-sealed structural/plumbing review for building permit acquisition, and a complete quantity takeoff and cost estimate.

Location & Hazard Context

This is not a neutral site. San Julian sits in the highest wind-speed corridor in the Philippines, within Seismic Zone 4, and on low-lying coastal terrain exposed to flooding and storm surge. Every major design decision on this project — the elevated +1.8m ground floor, the mat foundation, the roof anchorage detailing — traces back to this hazard profile, not just to the building's occupancy.

Building Type & Construction Method

Two-storey reinforced concrete frame residence with non-load-bearing CHB infill walls, cast-in-place mat foundation, engineered fill to +1.8m for flood elevation, steel truss and purlin roof system, and an attached in-ground swimming pool.

Client Objectives

The client required:

- ✓ Structural review and sign-off for building permit acquisition
- ✓ Permit-ready drawing revisions addressing code and site-hazard gaps
- ✓ A complete, verifiable quantity takeoff
- ✓ A detailed cost estimate backed by DUPA
- ✓ An organized Bill of Quantities
- ✓ Design recommendations appropriate to a flood-prone, seismic, coastal site
- ✓ Full technical documentation to support both permitting and construction

My Responsibilities

This is important: this section is about my work, not the tools I used to do it faster.

- ✓ Reviewing architectural, structural, and plumbing drawings
- ✓ Identifying missing or inconsistent information across the drawing set
- ✓ Reviewing structural and geotechnical adequacy against NSCP 2015 and the site's hazard profile
- ✓ Developing and logging RFIs
- ✓ Performing quantity takeoffs using Bluebeam and Excel
- ✓ Preparing the Bill of Quantities and DUPA
- ✓ Preparing the detailed cost estimate
- ✓ Exercising engineering judgment on assumptions where field data (soil investigation, supplier quotes) was unavailable
- ✓ Technical reporting and full documentation

One clear boundary: the electrical quantity takeoff was completed with AI assistance since electrical work sits outside my core Civil Engineering and Master Plumber licensure — it is flagged in the estimate for review by a licensed Professional Electrical Engineer before final client submission. Knowing the edge of your own scope is part of the job.

Software Used

Bluebeam Revu

Measurement, markup, drawing review, and revision tracking — area, count, and length tools with color-coded layers for traceability.

Microsoft Excel

Detailed quantity takeoff, Bill of Quantities, DUPA (DPWH format), cost estimate, and the material rate/rental reference database built up from past projects.

AutoCAD

Cross-reference for drawing review where vector detail was needed beyond what the PDF markup allowed.

Claude

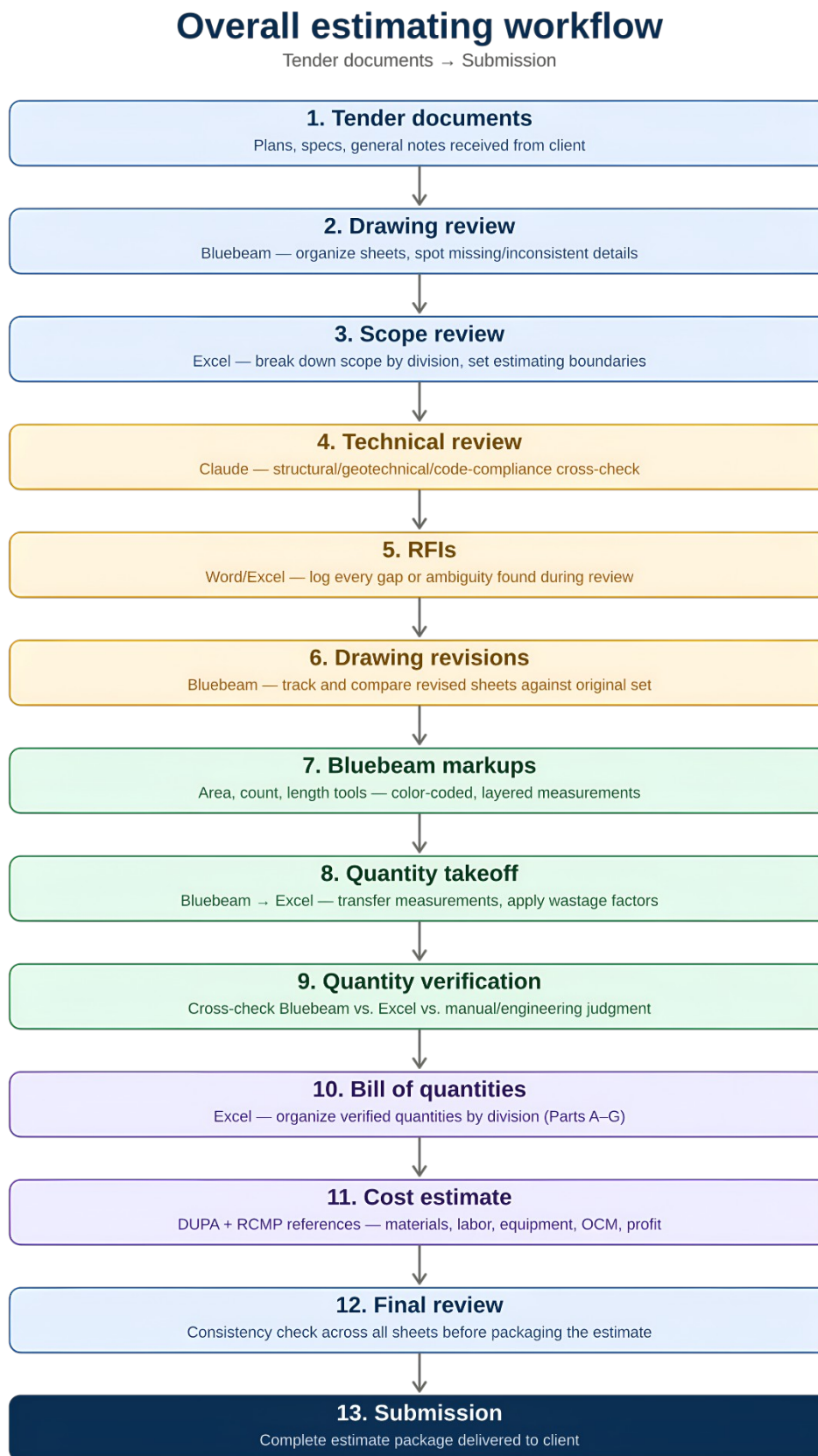
Structural and geotechnical cross-checking against NSCP 2015, since no structural analysis had originally been provided for this project. Used to pressure-test my own engineering judgment against the code — not to replace it.

ChatGPT

Workflow organization and documentation structuring.

Every engineering decision, revision, and final number in this package is mine. AI tools were used to verify against code, check my assumptions, and speed up documentation — the sign-off is mine, and so is the responsibility.

Overall Estimating Workflow



Review phase (blue) · Technical/AI-assisted phase (amber) · Measurement phase (green) · Estimating phase (purple)

Tender Document Review

The full drawing set was reviewed before any measurement began: cover sheet and general notes, the complete architectural set (A.01–A.14), structural set (S.01–S.14), plumbing set (P.01–P.07), and electrical set (E.01–E.04).

What I checked

- Stated occupancy classification and load assumptions in the general notes
- Foundation type and the soil bearing pressure assumed in the structural notes
- Consistency between architectural and structural dimensions across sheets
- Completeness of schedules (doors/windows, ceiling finishes, lintel/reinforcement tables)

How I organize drawings

Grouped by discipline, cross-indexed against the sheet's own table of contents, with every sheet that referenced a detail not actually included in the set flagged immediately — for example, the plumbing set repeatedly references a “Swimming Pool Design & Specifications” document that had to be tracked down and confirmed separately.

Scope Review and Summary

Before any measurement, the project scope was broken down by division to set clear estimating boundaries and reduce the risk of missed components — including the design review scope, which isn't usually part of a standard estimating engagement.

Division	Scope Summary	Value (₱)
A — General Requirements	Permits, OSH, mobilization/demobilization	67,255.00
B — Site Clearing & Disposal	Removal of existing structures/obstructions	21,671.44
C — Earthworks & Site Prep	Excavation, backfill/ground improvement, termite treatment, damp-proofing, gravel bedding	478,342.03
D — Structural Works	Concrete, reinforcement, formwork, steel truss & purlins	3,471,961.27
E — Architectural Works	Roofing, CHB walls, tiles, painting, ceiling, doors & windows	2,920,475.70
F — Plumbing Works	Fixtures, water lines, sewer, pool plumbing, storm, water storage	423,462.76
G — Electrical Works	Service entrance, wiring, lighting, outlets, pool electrical	203,063.32

Technical Engineering Review

This is the part of my process most estimators skip — and it's the part that mattered most on this project, since no structural analysis had been done before it reached me.

What I reviewed

- Structural adequacy — occupancy classification, load assumptions, seismic/wind parameters
- Geotechnical assumptions — assumed bearing capacity, foundation type given a coastal/flood-prone soil profile
- Constructability — whether the drawn details (e.g., grade beam vs. mat, roof anchorage) could actually be built as shown
- Code compliance — against NSCP 2015 and general Philippine building code practice
- Site-specific risk — flood elevation, typhoon wind exposure, seismic zone

Claude review summary (my curated takeaways, not the full report)

- Corrected an Occupancy Category I (“essential facility”) classification carried over from what was clearly a reused school-building template — this residence is Category IV
- Flagged an undocumented 340 kph design wind speed against the true NSCP 2015 figure for this region (≈250 kph) and had it reconciled
- Resolved a mismatch between the owner’s requested mat foundation and drawings that still showed isolated footings with grade beams
- Worked out the correct detailing for a slab-on-grade cast over engineered backfill on a flood-elevated site — including the isolation joint between the slab and the structural frame, which is the detail most commonly missed on this type of construction in the Philippines
- Flagged the original roof truss anchorage (10mmØ bolts, 300mm embedment) as light for this wind corridor and had it upsized

My decisions

I made the final call on every member size that ended up in the executed drawings. The owner wanted a mat foundation, and I agreed with that decision on my own engineering judgment given the coastal, flood-prone soil profile — the AI review gave me a code-referenced basis to act on, but the sign-off and the responsibility are mine.

Requests for Information (RFIs)

#	Question / Item	Reason	Status
1	Pool backwash drainage — tie-in point not shown on P.06	Pool spec prohibits tying backwash directly into the septic system	Open
2	Gutter downspout hardware (end caps, outlets, brackets) not shown	Needed for a complete, functional storm drainage system	Open
3	Circuit tag C-08 appears on both E.03 and E.04	Confirm if this is one continuous circuit across floors or a labeling error	Open
4	‘SOLAR’ lighting labels (17 pts, Ground Floor) — fixture type undefined	No fixture type/spec given anywhere in the electrical set	Open
5	Lighting fixture schedule not provided	Required for accurate pricing (type, wattage, lamp/LED spec, mounting height)	Open
6	Receptacle symbol counts on E.03/E.04 not legible in supplied PDF	Affects accuracy of the power/outlet quantity takeoff	Open
7	Weatherproof GFCI outlet at pool deck not shown on any E-sheet	Code-required near pool installations	Open
8	No electrical drawing for pool equipment (pump/filter/chlorinator)	Equipment shown on plumbing layout (P.06) has no electrical counterpart	Open
9	Window W5 listed in schedule/estimate but not located on any floor plan or elevation	Cannot verify quantity or lintel requirement without a drawn opening	Open — see Addendum Memo
10	No railing detail found for the stair or the second floor balcony	Life-safety item; confirm if a detail exists outside this set or needs to be designed	Open — see Addendum Memo

Drawing Revisions

Revisions were tracked against the original issued set using Bluebeam’s markup and comparison tools. The table below summarizes the structural revisions that were actually adopted into the executed drawings, cross-checked against the final QTO workbook.

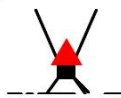
Item	Original	Revised
Mat foundation thickness	Not clearly specified / assumed thin	400mm — governed by punching shear check
Mat reinforcement	25mmØ @ 110mm o.c. (congested)	16mmØ @ 150mm o.c., both ways, both faces
Ground level structural element	Tie beam	Grade beam, appearing only above the +1.8m backfill level
Slab-on-grade (SOG) thickness	Unspecified	125mm, confirmed final
Columns	Mixed sizing across sheets	400×400mm, uniform ground floor & 2nd floor
Beams (GTB / 2F)	300×500 / 300×450mm (early proposal)	300×400mm, uniform, as adopted in final drawings
Roof truss base anchorage	4-10mmØ A325 bolts, 300mm embedment	4-16mmØ A325 bolts, 400mm embedment

Bluebeam Markups

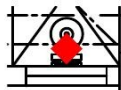
Construction of 2 Storey Residential Building with pool.pdf Markup Summary 01-08-2026

Plumbing Fixtures (10)

Count: 10 Measurement: 10 Count



Subject: Plumbing Fixtures
Length: 0
Area: 0.00
Wall Area: 0
Volume: 0
Count: 1
Measurement: 1 Count
Label: Floor Drain
Layer: Plumbing



Subject: Plumbing Fixtures
Length: 0
Area: 0.00
Wall Area: 0
Volume: 0
Count: 1
Measurement: 1 Count
Label: Shower Head
Layer: Plumbing



Subject: Plumbing Fixtures
Length: 0
Area: 0.00
Wall Area: 0
Volume: 0
Count: 1
Measurement: 1 Count
Label: Lavatory
Layer: Plumbing



Subject: Plumbing Fixtures
Length: 0
Area: 0.00
Wall Area: 0
Volume: 0
Count: 1
Measurement: 1 Count
Label: Water Closet
Layer: Plumbing

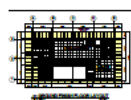


Subject: Plumbing Fixtures
Length: 0
Area: 0.00
Wall Area: 0
Volume: 0
Count: 1
Measurement: 1 Count
Label: Kitchen Sink
Layer: Plumbing

Floor Finish (5)
Length: 0.00 m **Area:** 125.08 sq m **Wall Area:** 4.62 sq m **Volume:** 4.43 cu m **Measurement:** 125.08 sq m



Subject: Floor Finish
Length: 15.40 m
Area: 14.78 sq m
Wall Area: 4.62 sq m
Volume: 4.43 cu m
Count: 0
Measurement: 14.78 sq m
Label: 600mm x 600mm UnGlazed Homogenous Tiles in
 matte finish
Layer: Architectural



Subject: Floor Finish
Length: 107073.00 mm
Area: 59.97 sq m
Wall Area: 0
Volume: 0
Count: 0
Measurement: 59.97 sq m
Label: Pavement Stones
Layer: Architectural

Sample n

Tools used

- Area Tool — slab, wall, and finish quantities (e.g., mat foundation footprint measured at 95.29 sqm)
- Count Tool — fixtures, doors/windows, lighting points
- Length Tool — CHB wall runs, piping runs
- Notes & Layers — every markup tagged to a discipline layer (Plumbing, Architectural, Structural, To Revise) with color coding for traceability

This page should demonstrate real Bluebeam fluency — not just that measurements were taken, but that they were organized so any reviewer could retrace exactly where a number came from.

Quantity Takeoff

Example — mat foundation concrete volume, traced end to end:

Step	Detail
1. Measured in Bluebeam	Mat foundation footprint area tool → 95.29 sqm
2. Transferred to Excel	Area carried into the FINAL BACKUP (rect) sheet
3. Verified	Cross-checked footprint against the structural mat foundation plan
4. Design thickness applied	0.4m confirmed thickness (governed by punching shear)
5. Result	95.25 sqm × 0.4m = 38.116 m³ concrete → carried to BOQ

Quantity Verification

This step is missing from most estimating portfolios — here's how it was actually done on this project:

- Cross-check between Bluebeam-measured areas/lengths and the Excel takeoff sheet, line by line
- Manual spot-checks against dimensioned callouts on the structural sheets (not scaled measurements alone)
- Engineering judgment applied wherever a drawing was ambiguous or incomplete — documented as an assumption, never silently guessed
- Every AI-assisted takeoff item (electrical scope) was individually re-checked by hand before being accepted into the final BOQ

Detailed Quantity Takeoff — Selected Highlights

Full detail lives in the appendix; the items below are the ones worth walking through.

Item	Basis	Quantity
Mat foundation concrete	95.25 sqm × 0.4m	38.116 m ³
Mat edge thickening	Perimeter × 0.4m × 0.6m	12.629 m ³
Slab-on-grade (SOG)	0.125m thickness	10.646 m ³
2nd floor slab	0.125m thickness	10.646 m ³
Columns (concrete)	400×400mm, ground & 2F	15.936 m ³
Reinforcing steel, Grade 60 (main bars)	Columns, tie beams, beams, lintel beams, slab, footings	10,977.41 kg
Reinforcing steel, Grade 40 (ties/stirrups)	All transverse reinforcement	4,979.15 kg
Formworks and falseworks	All cast-in-place elements	183.34 sqm
Engineered select granular backfill (to +1.8m)	Platform area net of pool, net of mat/gravel/blinding	364.5 m ³
Surcharge preload fill (temporary)	Ground improvement per Structural Review Sec. 12	≈100.0 m ³

Bill of Quantities — Summary

Part	Division	Total (P)
A	General Requirements	67,255.00
B	Site Clearing, Demolition & Disposal	21,671.44
C	Earthworks & Site Preparation	478,342.03
D	Structural Works	3,471,961.27
E	Architectural Works	2,920,475.70
F	Plumbing Works	423,462.76
G	Electrical Works	203,063.32
	TOTAL PROJECT COST	7,586,231.52

Structural (Part D) and Architectural (Part E) works together make up roughly 85% of total project cost — consistent with a hazard-driven design where foundation, frame, and envelope carry most of the budget.

Detailed Cost Estimate

Component	Amount (P)
Materials	4,654,690.86
Labor	865,399.28
Equipment	585,104.92
Total Direct Cost	6,105,195.05
OCM (Overhead, Contingency & Miscellaneous)	905,823.01
Contractor's Profit	483,985.60

Component	Amount (₱)
TOTAL PROJECT COST (VAT-inclusive)	7,586,231.52

Pricing basis: historical prices from my past projects, DPWH RCMP (Regional Consolidated Market Prices), DPWH DUPA format for unit price analysis, and engineering judgment where current supplier data wasn't yet available. Future revisions will replace placeholder pricing with actual supplier quotations.

Supplier Validation — Planned Next Step

Not yet completed on this engagement — flagged here as a deliberate next step for future estimates, since current pricing is still reference-based rather than quoted.

Item	Supplier A	Supplier B	Supplier C
Ready-mix concrete, f'c=27.6 MPa	To be quoted	To be quoted	To be quoted
Reinforcing steel, Grade 60	To be quoted	To be quoted	To be quoted
PPGI roofing sheet, 0.6mm	To be quoted	To be quoted	To be quoted

Building this comparison table into the estimating workflow — rather than treating supplier validation as an afterthought — is one of the maturity improvements I'm carrying into future engagements.

Risks Identified

No soil investigation was conducted — all bearing capacity values in the structural notes are assumed, not tested.

Flood-prone coastal site — required a full re-think of the ground floor system (elevated slab-on-grade over engineered backfill), not just a foundation upsize.

Seismic Zone 4 — highest seismic zone designation nationally.

No structural analysis was originally provided — initial member sizes were dimension-only, not calculated.

Drawing inconsistencies across revisions — required active version tracking to avoid pricing an outdated detail.

Missing specifications — lighting fixture schedule, pool equipment electrical layout, and other items logged as RFIs.

Placeholder material brands/allowances used pending client/supplier confirmation.

Challenges

Organizing a large, inconsistent drawing set into a structured estimating workflow was the first challenge — but not the hardest one.

The harder challenge was structural: the design showed a lot of details that weren't standard code-compliant practice, and no structural analysis had been performed — member sizes came from initial dimensioning, not calculation. I offered the client a proper structural analysis and a soil investigation for the mat foundation design; both were declined. I couldn't force either one, but I still had to sign and seal the plans. I had already sunk hours into this project before that became clear, and backing out at that point meant walking away unpaid for work already done — so I stayed and faced it directly, using AI tools for checking and verification rather than walking away.

The site itself compounded this: it's genuinely earthquake-prone and sits in a heavy typhoon corridor along the Pacific side of Samar, and it's coastal. Given all of that, an isolated footing was the wrong call — the owner independently wanted a mat foundation, which I agreed with, and I redesigned the other structural members around that decision with Claude's help to check my reasoning against the code.

The +1.8m flood elevation added its own layer: elevating soil like that properly needs an engineered fill methodology and a retaining structure to hold the elevated soil in place while also supporting the structure above it — not just a taller foundation.

On top of the structural struggles, the drawing set itself had a lot of errors, and the resulting revisions repeatedly delayed my quantity takeoff and scope assessment. Keeping Bluebeam measurements and Excel calculations consistent while all of this was moving took real discipline, as did managing assumptions where supplier pricing wasn't yet available.

Solutions

- Built a structured estimating workflow from drawing review through to final estimate preparation, instead of measuring ad hoc.
- Used Bluebeam markups together with Excel verification to reduce omissions and keep quantities traceable.
- Logged every note and revision, marking up drawings in Bluebeam so revisions were clearly spotted and identified.
- Organized estimating documents into clearly defined sections for easier review and future updates.
- Planned to replace preliminary pricing with supplier quotations to improve estimate reliability going forward.
- Used AI deliberately — not only to work faster, but for critical thinking, cross-checking against code, and catching things I might otherwise have missed. I verified every AI-assisted output one by one, correcting, adding missing scope, and fixing wrong takeoffs myself before anything went into the final package.

Final Deliverables

- ✓ Drawing Review
- ✓ Bluebeam Markups
- ✓ RFI Log
- ✓ Quantity Takeoff
- ✓ Bill of Quantities
- ✓ Detailed Cost Estimate
- ✓ Technical Review Report
- ✓ Design Recommendations & Addendum

Key Results

Metric	Result
Structural gaps caught before permitting	Occupancy classification, wind speed basis, foundation system, roof anchorage — all corrected before sign-off
Estimating omissions reduced	10 RFIs logged and tracked, including 2 gap items (W5, railings) resolved via formal addendum
Documentation	Traceable quantities — every BOQ figure can be walked back to a Bluebeam measurement or a structural calculation
Turnaround	Complete estimate package delivered within 48 hours of engagement
Total Project Cost	₱7,586,231.52, fully broken down by division and by material/labor/equipment/markup

Lessons Learned

- A structured workflow reduces estimating errors.
- Clear client communication and boundary-setting — including a proper contract agreement — protects both sides.
- Clear documentation improves review and collaboration.
- Quantity verification is just as important as the measurement itself.
- Reliable pricing should be backed by current supplier quotations whenever possible.
- Continuous improvement in both technical skill and estimating systems leads to better project outcomes.
- Before accepting a project, check the tendered documents first — completeness, complexity, design risk, and level of detail all belong in the rate, not just the hours.

Reflection

This project strengthened my understanding of how a systematic estimating workflow improves both accuracy and efficiency. Beyond preparing quantities, I learned the importance of organizing documentation, validating measurements, and maintaining traceable records throughout the process. Going forward, I plan to incorporate actual supplier quotations, improve cost validation, and keep refining my workflow through better tools and better habits.

I'll say this plainly: clients should get their plans done by professionals to begin with, to avoid the volume of revisions this project needed — not just for aesthetics, but because safety has to come first. I may not have earned as much from this project as the effort deserved, but it taught me a lot. Going forward, before accepting a rate, I'll check the tendered documents first — their completeness, their complexity, the design risks, the extent of detail involved. I may ask for a downpayment upfront, but the final rate will reflect the hours actually worked, the scope actually completed, or the overall complexity of the project as a lump-sum figure, regardless of how long it takes.

A contract agreement is necessary to protect both of us. I don't want to take advantage of my clients — I just want to work and be compensated fairly. The client gets quality output; I get the compensation my service and effort deserve.

Appendix

Full supporting documentation is maintained separately and available on request. This case study intentionally keeps the main narrative focused — the appendix is where completeness lives.

Document	Contents
Full Bill of Quantities	Complete line-item BOQ, all divisions (Parts A–G)
Complete Excel Workbook	SOW, BOM/BOQ, estimate, DUPA, POW, manpower & equipment, material rates — full computation backup
Complete Bluebeam Markups	Full markup summary, all disciplines, all sheets
Complete RFI Log	All 10 logged items with full remarks and cross-references
Structural & Geotechnical Review Report	Full technical review with code basis, revised general notes, and structural member schedule
Design & Estimate Addendum Memo	W5 window and railings follow-up, per personal review notes
Swimming Pool Design & Specifications	Referenced pool design/spec document
Full Drawing Set	Architectural, structural, plumbing, and electrical sheets, as revised

SELECTED DOCUMENTATION & APPENDIX

Construction of a Two-Storey Residential Building with Swimming Pool

Samar, Philippines

This document supports the main case study for the Construction of a two-storey Residential Building. It contains selected proof-of-work excerpts from the estimating workbook, drawing set, and Bluebeam markup file, followed by one worked quality-control example and a full appendix index of supporting documents available on request.

Selected Documentation

Drawing Set



01
A.01 / SCALE: N.T.S.

PERSPECTIVE

02
A.01 / SCALE: M.T.S.

LOCATION MAP



THIS SITE

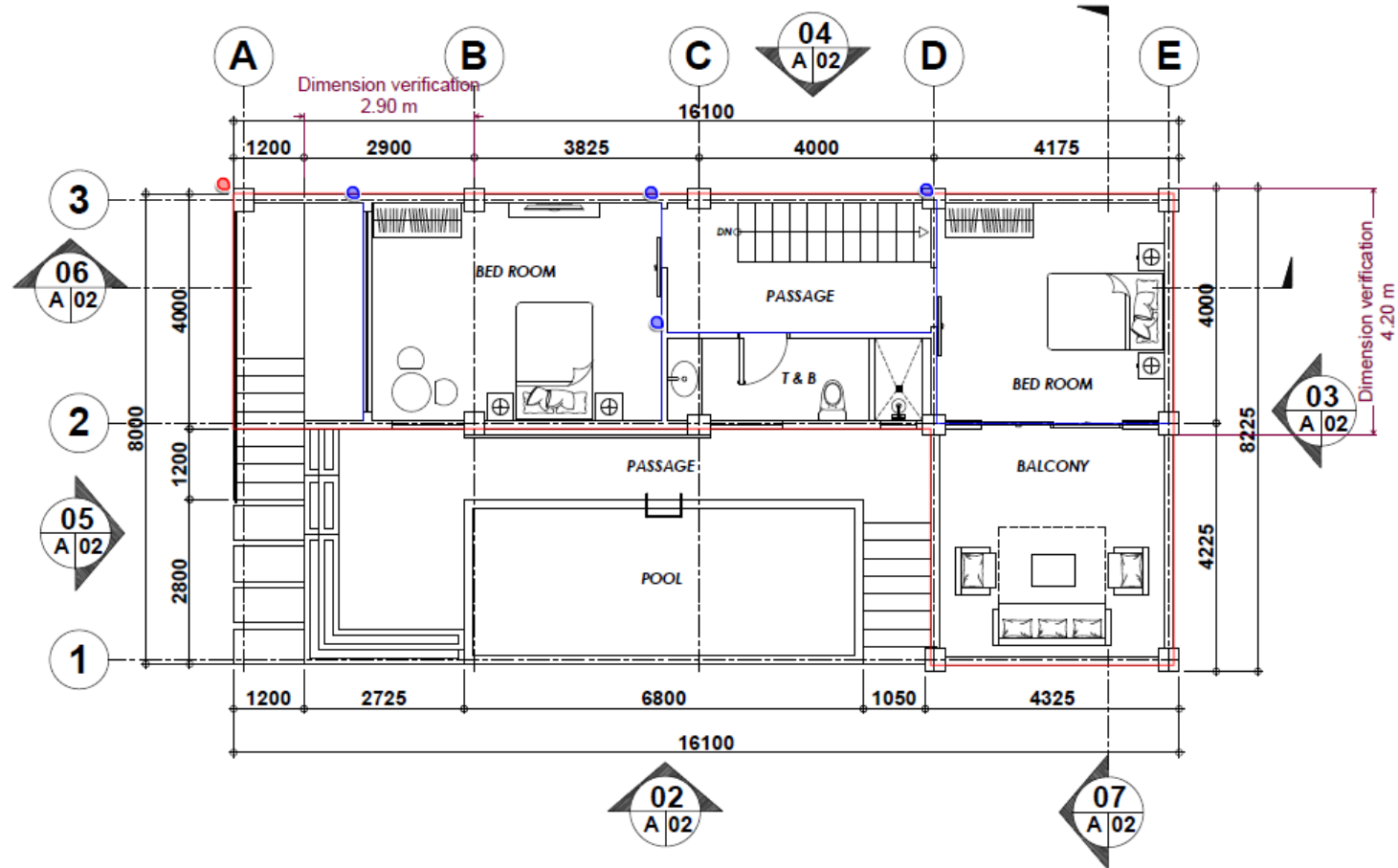
TABLE OF CONTENTS	
ARCHITECTURAL	
A.01 - PERSPECTIVE LOCATION MAP	
A.02 - GROUND FLOOR PLAN	
A.03 - SECOND FLOOR PLAN	
A.04 - ROOF PLAN	
A.05 - FRONT ELEVATION	
A.06 - RIGHT SIDE ELEVATION & LEFT SIDE ELEVATION	
A.07 - REAR ELEVATION	
A.08 - LONGITUDINAL SECTION	
A.09 - CROSS SECTION	
A.10 - GROUND FINISH FLOOR LAYOUT	
A.11 - SECOND FINISH FLOOR LAYOUT	
A.12 - GROUND FLOOR REFLECTED CEILING PLAN	
A.13 - SECOND FLOOR REFLECTED CEILING PLAN	
A.14 - SCHEDULE OF DOORS AND WINDOWS	
STRUCTURAL	
S.01-S.03 - GENERAL CONSTRUCTION NOTES	
S.04 - FOUNDATION PLAN	
S.05 - GRADE BEAM DETAILED LAYOUT	
S.06 - FLOOR BEAM DETAILED LAYOUT	
S.07 - ROOF/TIE BEAM LAYOUT DETAILED LAYOUT	
S.08 - STAIRCASE DETAIL AND REINFORCEMENT	
S.09 - MAT FOUNDATION SECTION	
S.10 - BEAM DETAIL AND COLUMN DETAIL	
S.11 - WALL FOOTING DETAIL	
S.12 - STEEL TRUSS DETAIL AND PURLINS DETAIL	
S.13 - ROOF FRAMING DETAIL	
S.14 - POOL STRUCTURAL DETAIL	
ELECTRICAL	
E.01 - GROUND FLOOR LIGHTING LAYOUT	
E.02 - SECOND FLOOR LIGHTING LAYOUT	
E.03 - GROUND FLOOR POWER LAYOUT	
E.04 - SECOND FLOOR POWER LAYOUT	
MECHANICAL	
FIRE PROTECTION	
PLUMBING & SANITARY	
P.01 - GROUND FLOOR SANITARY LAYOUT	
P.02 - GROUND FLOOR SANITARY AND WASTE WATER LINE LAYOUT	
P.03 - SECOND FLOOR SANITARY AND WASTE WATER LINE LAYOUT	
P.04 - ROOF PLUMBING LAYOUT	
P.05 - SECOND FLOOR WATER LINE LAYOUT	
P.06 - POOL PLUMBING LAYOUT	
P.07 - SANITARY AND PIPING DETAILS	

PROJECT AND LOCATION:
CONSTRUCTION OF TWO-STOREY
RESIDENTIAL BUILDING WITH POOL

SHEET CONTENT:
PERSPECTIVE
LOCATION MAP

SHEET NO.:
A
01/16

Sheet A.01 — Perspective and location map, as issued in the drawing set

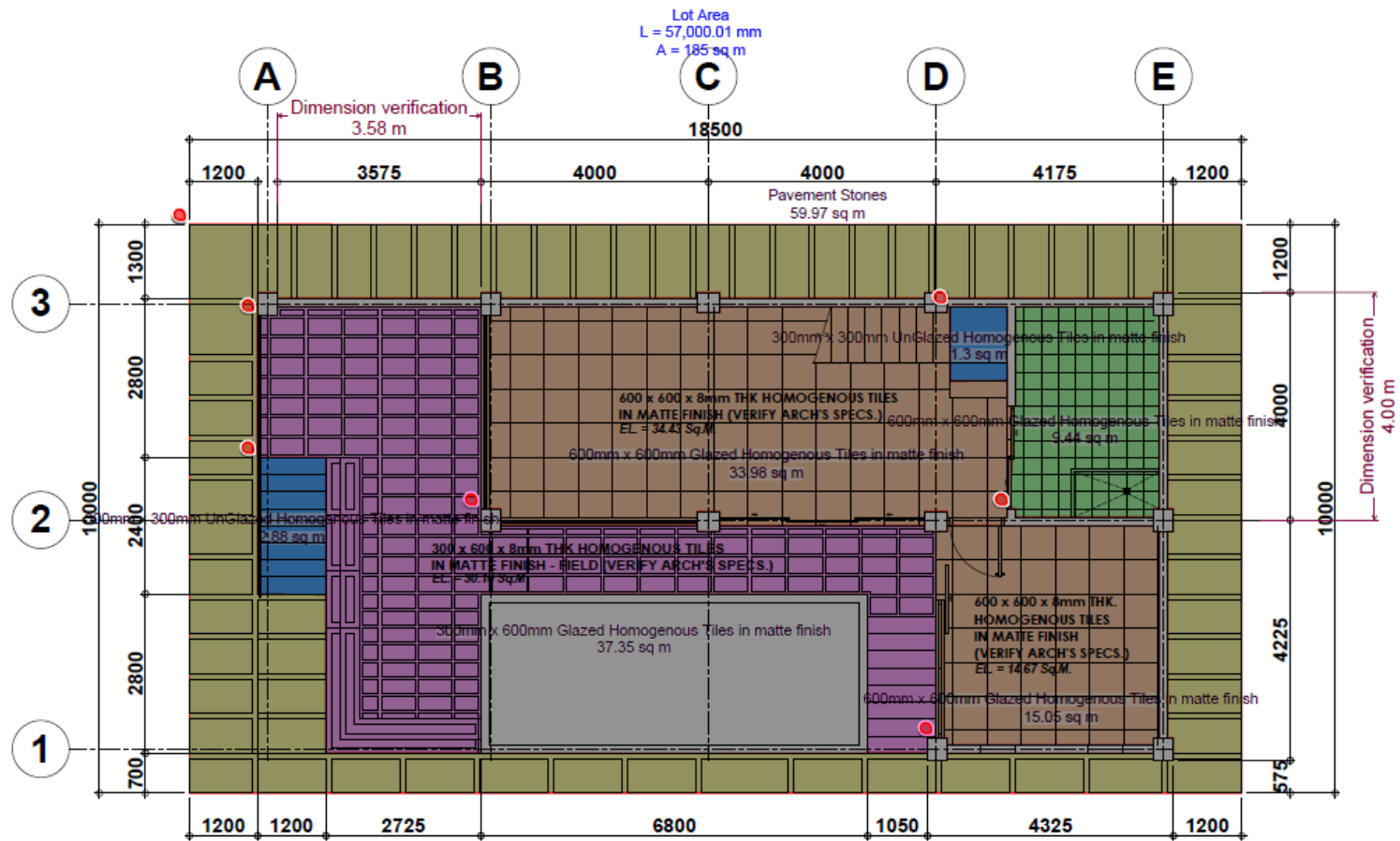


01 SECOND FLOOR PLAN

A 03 SCALE 1:75 MTS.

	PROJECT AND LOCATION: CONSTRUCTION OF TWO-STOREY RESIDENTIAL BUILDING WITH POOL		SHEET CONTENT: SECOND FLOOR PLAN	SHEET NO.: A 03/16
--	---	--	-------------------------------------	--------------------------

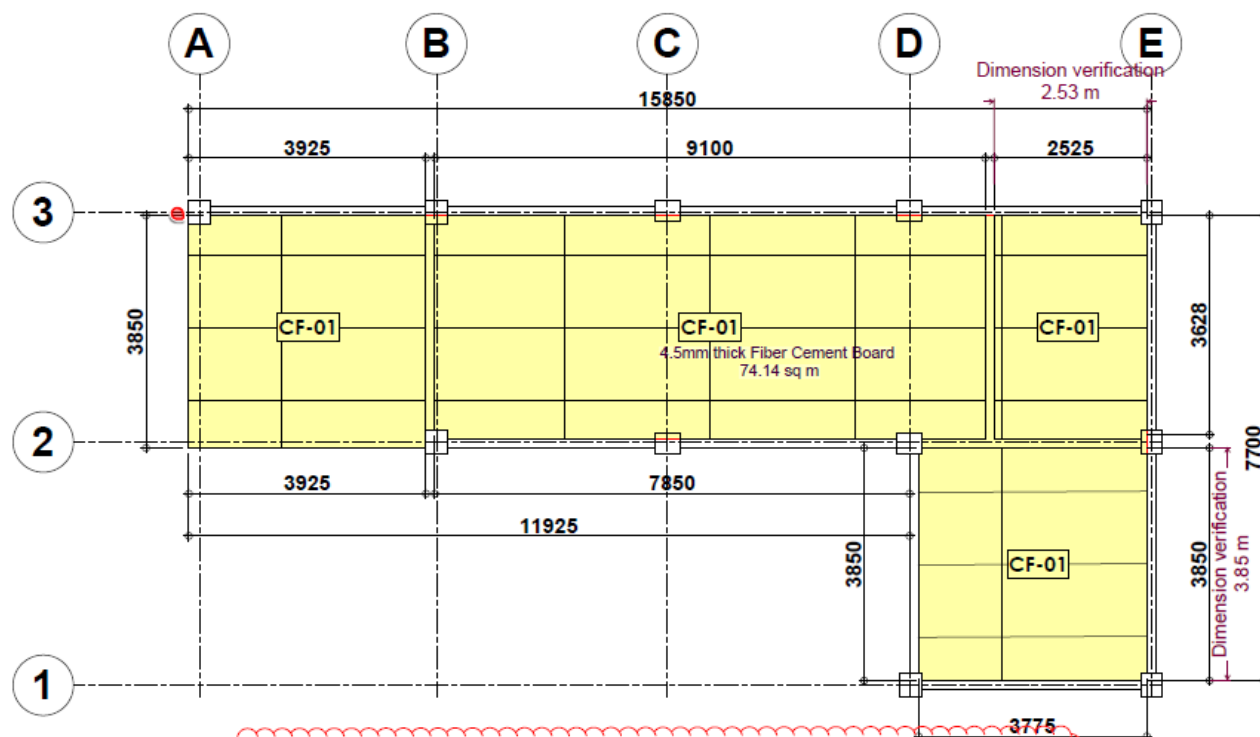
Sheet A.03 — Second Floor plan with Bluebeam Length tool to find the Wall Area, Openings are deducted in excel computation



01 GROUND FINISH FLOOR LAYOUT

A/10 SCALE 1:75 MTS.

	PROJECT AND LOCATION: CONSTRUCTION OF TWO-STOREY RESIDENTIAL BUILDING WITH POOL		SHEET CONTENT: GROUND FINISH FLOOR LAYOUT	SHEET NO.: A 10/16
--	---	--	---	--------------------------



SCHEDULE OF CEILING FINISHES

FINISHES	DESCRIPTION
CF-01	12mm THK MOISTURE RESISTANT GYPSUM BOARD ON METAL FURRING; PAINTED FINISH (VERIFY ARCHITECT'S SPECS.)

Change Material to
4.5mm fiber Cement
Board

01
A 12

GROUND FLOOR REFLECTED CEILING PLAN

SCALE

1:75 MTS.

PROJECT AND LOCATION:

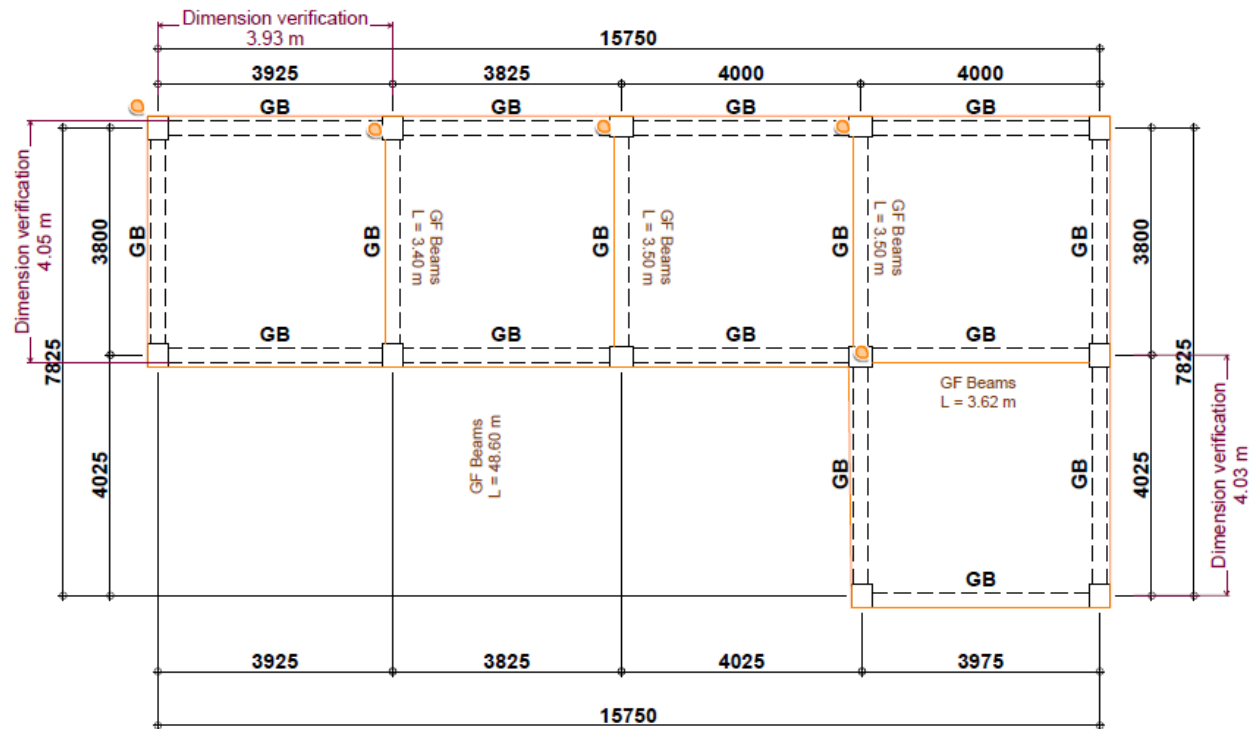
CONSTRUCTION OF TWO-STOREY
RESIDENTIAL BUILDING WITH POOL

SHEET CONTENT:

GROUND FLOOR
REFLECTED CEILING PLAN

SHEET NO.:

A
12/16

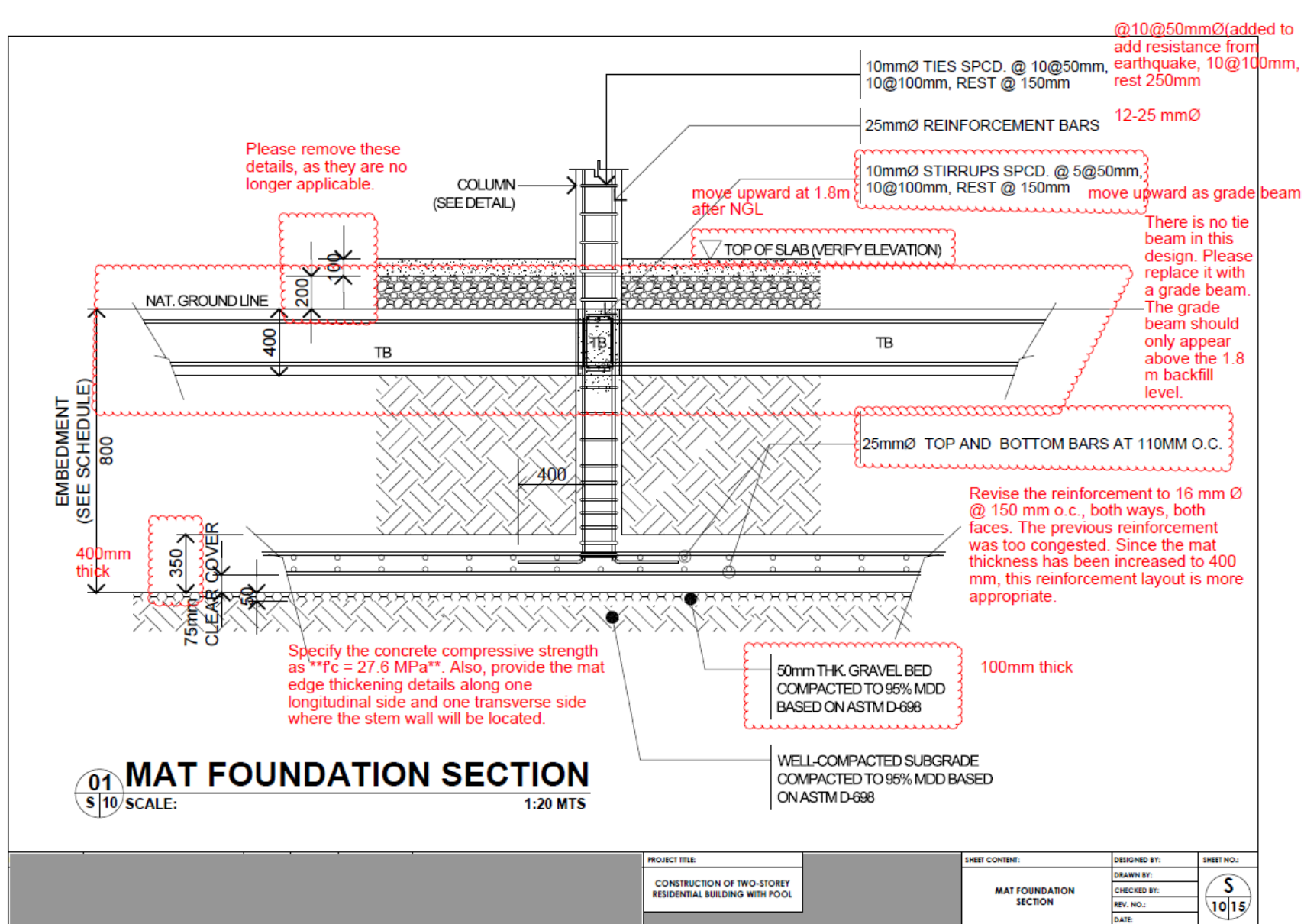


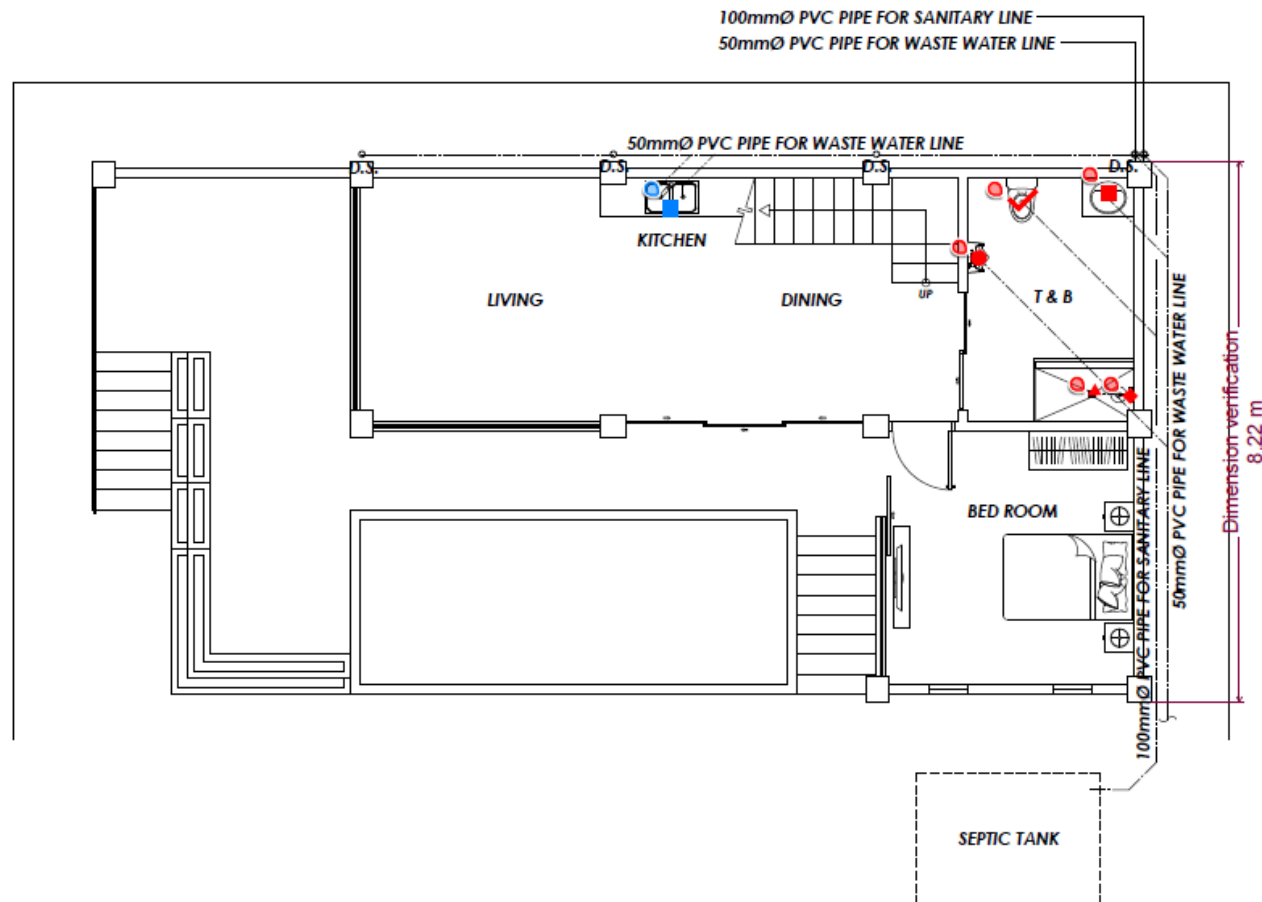
01 S 05

GRADE BEAM DETAILED LAYOUT

SCALE 1:75 MTS.

	PROJECT AND LOCATION:		SHEET CONTENT:	SHEET NO.:
	CONSTRUCTION OF TWO-STOREY RESIDENTIAL BUILDING WITH POOL		GRADE BEAM DETAILED LAYOUT	S 05/14





01
P 01

GROUND FLOOR SANITARY AND WASTE WATER LINE LAYOUT

SCALE

1:75 MTS.

PROJECT AND LOCATION:

CONSTRUCTION OF TWO-STOREY
RESIDENTIAL BUILDING WITH POOL

SHEET CONTENT:

GROUND FLOOR SANITARY
AND WASTE WATER LINE
LAYOUT

SHEET NO.:

P
01/07

Estimating Workbook

BILL OF MATERIALS/BILLOF QUANTITIES					
Name of Project: CONSTRUCTION OF TWO-STOREY RESIDENTIAL BUILDING WITH POOL					
Location: Baranqay 3 poblacion. Cancasayo Road San Julian E. Samar					
Estimated No. of Construction Days:				120.00	Calendar Days
Estimated Project Cost:					P 7,586,231.52
Item No.	DESCRIPTION	Qty	Unit	UNIT COST	TOTAL COST
PART A GENERAL REQUIREMENTS					
A.1	Permits and Clearances	1.00	l.s.		
A.2	Occupational Safety and Health	1.00	l.s.		
A.3	Mobilization/Demobilization	1.00	l.s.		
TOTAL OF PART A					
PART B SITE CLEARING, DEMOLITION & DISPOSAL					
B.1	Removal of structures and obstructions, including clearing and grubbing	1.00	l.s.		
TOTAL OF PART B					
PART C EARTHWORKS & SITE PREPARATION					
C.1	Structural excavation	100.00	cu.m.		
C.2	Backfill, Compaction, and Ground Improvement	465.00	cu.m.		
C.3	Soil Treatment/ Anti-Termite Soil Poisoning	87.17	L		
C.4	Cofferdaming/ Dewatering	1.00	lot		
C.5	Damproofing (Polyethylene Sheets, Pool Crystalline Waterproofing and PVC Waterstops)	1.00	lot		
C.6	Gravel Bedding	15.21	cu.m.		
TOTAL OF PART C					
PART D STRUCTURAL WORKS					
D.1	Concrete Works: Structural Concrete, Class AA , 4000 PSI (28 Days)	149.50	cu.m.		
D.2	Reinforcing Steel (Deformed), Grade 60 (Fy = 414 MPa), for main bars of columns, tie beams, beams, lintel beams, slab, PC1, and footings (PCC)	10,977.41	kg		
D.3	Formworks and Falseworks	183.34	sq.m.		
D.4	Reinforcing Steel (Deformed), Grade 40 (Fy = 276 MPa), for ties and stirrups	4,979.15	kg		
D.5	Structural Steel Truss	299.51	kg		
D.6	Structural Steel Purlins	427.35	kg		
D.7	Other Roof Materials and Accessories: Flashings, Screws, Blind rivets, PU Sealants, Sagrods, Roof Bracing, Base Plates and Anchor Bolts.	1.00	lot		

PART E ARCHITECTURAL WORKS					
E.1	Prepainted Metal Sheets (0.6mm) Rib Type Long Span	62.34	sq.m.		
E.2	150mm Non-Load Bearing CHB, including Reinforcing Steel and Plastering	348.18	sq.m.		
E.3	100mm Non-Load Bearing CHB, including Reinforcing Steel and Plastering	168.01	sq.m.		
E.4	Glazed Tiles	147.29	sq.m.		
E.5	Unglazed Tiles	138.28	sq.m.		
E.6	Painting Works	1,303.58	sq.m.		
E.7	2 Coats-Liquid-applied polyurethane waterproofing (CR, interior walls of the septic tank and cistern tank)	182.03	sqm		
E.9	Ceiling Works (Metal Frame, 4.5mm Fiber Cement)	150.30	sqm		
E.10	Doors and Windows	1.00	lot		
TOTAL OF PART E					
PART F PLUMBING WORKS					
F.1	Plumbing Fixtures	1.00	l.s.		
F.2	Fresh Water Lines and Pipes	1.00	l.s.		
F.3	Sewer and Sanitary Works	1.00	l.s.		
F.4	Pool Plumbing	1.00	l.s.		
F.5	Rainwater Storm	1.00	l.s.		
F.6	Water Storage	1.00	l.s.		
TOTAL OF PART F					
PART G ELECTRICAL WORKS					
G.1	Service Entrance, Panel, & Feeders	1.00	l.s.		
G.2	Branch wiring & Conduits	1.00	l.s.		
G.3	Lighting Fixtures and Switches	1.00	l.s.		
G.4	Power - Convenience Outlets	1.00	l.s.		
G.5	Grounding and Bonding	1.00	l.s.		
G.6	Pool Electrical	1.00	l.s.		
G.7	Boxes, Fittings and permits	1.00	l.s.		
TOTAL OF PART G					
Total Project Cost					7,586,231.52
Prepared by:					
ENGR. MA. JESSA P. DOLLADO, RMP.					
<i>Civil Engineer</i>					

Bill of Materials/Quantities Format

DETAILED COST ESTIMATE													
Name of Project: CONSTRUCTION OF TWO-STOREY RESIDENTIAL BUILDING WITH POOL							Estimated No. of Construction Days: 120 Calendar Days						
Location: Barangay 3 poblacion, Cancasayo Road San Julian E. Samar							Estimated Project Cost: ₱ 7,586,231.52						
Item No.	DESCRIPTION	Qty	Unit	DIRECT COST			TOTAL DIRECT COST	MARK-UP		TOTAL MARK-UP	VAT	Unit Cost	Total Cost
				MATERIALS	LABOR	EQUIPMENT		OCM	PROFIT				
PART A	GENERAL REQUIREMENTS												
A.1	Permits and Clearances	1.00	I.s.	35,000.00	0.00	0.00	35,000.00	0.00	0.00	0.00		35,000.00	35,000.00
A.2	Occupational Safety and Health	1.00	I.s.	11,000.00	0.00	0.00	11,000.00	0.00	880.00	880.00		11,880.00	11,880.00
A.3	Mobilization/Demobilization	1.00	I.s.	0.00	0.00	20,375.00	20,375.00	0.00	0.00	0.00		20,375.00	20,375.00
TOTAL OF PART A				46,000.00	-	20,375.00	66,375.00	-	880.00	880.00	-		67,255.00
PART B	SITE CLEARING, DEMOLITION & DISPOSAL												
B.1	Removal of structures and obstructions, including clearing and grubbing	1.00	I.s.	15,048.75	2,336.64	233.66	17,619.05	2,642.86	1,409.52	4,052.38		21,671.44	21,671.44
TOTAL OF PART B				15,048.75	2,336.64	233.66	17,619.05	2,642.86	1,409.52	4,052.38	-		21,671.44
PART C	EARTHWORKS & SITE PREPARATION												
C.1	Structural excavation	100.00	cu.m.	0.00	2,427.62	31,146.67	33,574.30	5,036.14	2,685.94	7,722.09		412.94	41,296.39
C.2	Backfill, Compaction, and Ground Improvement	465.00	cu.m.	127,575.00	16,181.14	106,408.67	250,164.80	37,524.72	20,013.18	57,537.90		661.73	307,702.71
C.3	Soil Treatment/ Anti-Termite Soil Poisoning	87.17	L	27,894.40	5,869.83	586.98	34,351.21	5,152.68	2,748.10	7,900.78		484.71	42,251.99
C.4	Cofferdaming/ Dewatering	1.00	lot	3,000.00	4,783.83	478.38	8,262.21	1,239.33	660.98	1,900.31		10,162.52	10,162.52
C.5	Dampproofing (Polyethylene Sheets, Pool Crystalline Waterproofing and PVC Waterstops)	1.00	lot	25,395.95	2,336.64	233.66	27,966.25	4,194.94	2,237.30	6,432.24		34,398.49	34,398.49
C.6	Gravel Bedding	15.21	cu.m.	28,750.68	4,340.66	1,485.84	34,577.18	5,186.58	2,766.17	7,952.75		2,795.81	42,529.93
TOTAL OF PART C				212,616.03	35,939.73	140,340.21	388,895.96	58,334.39	31,111.68	89,446.07	-		478,342.03
PART D	STRUCTURAL WORKS												
D.1	Concrete Works: Structural Concrete, Class AA , 4000 PSI (28 Days)	149.50	cu.m.	1,177,330.46	29,842.28	91,926.58	1,299,099.33	194,864.90	103,927.95	298,792.85		10,688.08	1,597,892.17
D.2	Reinforcing Steel (Deformed), Grade 60 (Fy = 414 MPa), for main bars of columns, tie beams, beams,lintel beams, slab, PC1, and footings (PCC)	10,977.41	kg	784,061.51	107,531.64	22,995.12	914,588.27	137,188.24	73,167.06	210,355.30		102.48	1,124,943.58
D.3	Formworks and Falseworks	183.34	sq.m.	39,755.54	50,519.70	3,780.75	94,056.00	14,108.40	7,524.48	21,632.88		631.02	115,688.87
D.4	Reinforcing Steel (Deformed), Grade 40 (Fy = 276 MPa), for ties and stirrups	4,979.15	kg	330,740.29	48,774.40	10,430.17	389,944.85	58,491.73	31,195.59	89,687.32		96.33	479,632.17
D.5	Structural Steel Truss	299.51	kg	24,980.99	2,459.12	8,874.22	36,314.33	5,447.15	2,905.15	8,352.29		149.13	44,666.62
D.6	Structural Steel Purlins	427.35	kg	36,413.43	2,832.30	6,716.78	45,962.51	6,894.38	3,677.00	10,571.38		132.29	56,533.88
D.7	Other Roof Materials and Accessories: Flashings,Screws, Blind rivets, PU Sealants, Sagrods, Roof Bracing, Base Plates and Anchor Bolts.	1.00	lot	42,446.18	292.08	29.21	42,767.46	6,415.12	3,421.40	9,836.52		52,603.98	52,603.98
TOTAL OF PART D				2,393,282.22	241,959.44	144,723.61	2,779,965.28	416,994.79	222,397.22	639,392.01	-		3,471,961.27
PART E	ARCHITECTURAL WORKS												
E.1	Prepainted Metal Sheets (0.6mm) Rib Type Long Soan	62.34	sq.m.	47,381.97	8,187.08	818.71	56,387.76	8,458.16	4,511.02	12,969.18		1,112.48	69,356.94
E.2	150mm Non-Load Bearing CHB, including Reinforcing Steel and Plastering	348.18	sq.m.	464,659.56	57,801.89	163,445.23	685,906.67	102,886.00	54,872.53	157,758.53		2,423.09	843,665.21
E.3	100mm Non-Load Bearing CHB, including Reinforcing Steel and Plastering	168.01	sq.m.	171,799.24	23,188.59	65,569.91	260,557.75	39,083.66	20,844.62	59,928.28		1,907.53	320,486.03
E.4	Glazed Tiles	147.29	sq.m.	129,979.74	102,430.75	10,243.08	242,653.57	36,398.04	19,412.29	55,810.32		2,026.37	298,463.89
E.5	Unglazed Tiles	138.28	sq.m.	116,265.73	67,313.36	6,731.34	190,310.42	28,546.56	15,224.83	43,771.40		1,692.86	234,081.82
E.6	Painting Works	1,303.58	sq.m.	147,689.30	238,886.80	23,888.68	410,464.79	61,569.72	32,837.18	94,406.90		387.30	504,871.69
E.7	2 Coats-Liquid-applied polyurethane waterproofing (CR, interior walls of the septic tank and cistern tank)	182.03	sqm	20,352.56	9,211.44	921.14	30,485.14	4,572.77	2,438.81	7,011.58		205.99	37,496.72
E.9	Ceiling Works (Metal Frame,4.5mm Fiber Cement)	150.30	sqm	154,431.35	43,969.51	4,396.95	202,797.81	30,419.67	16,223.82	46,643.50		1,659.58	
E.10	Doors and Windows	1.00	lot	497,204.40	363.62	36.36	497,604.39	74,640.66	39,808.35	114,449.01		612,053.39	612,053.39
TOTAL OF PART E				1,740,763.85	551,353.05	276,051.39	2,577,169.30	386,675.24	206,473.46	593,148.71			2,920,475.70

PART F	PLUMBING WORKS												
F.1	Plumbing Fixtures	1.00	I.s.	47,700.00	5,817.98	581.80	54,099.78	8,114.97	4,327.98	12,442.95		66,542.73	66,542.73
F.2	Fresh Water Lines and Pipes	1.00	I.s.	22,100.00	5,817.98	581.80	28,499.78	4,274.97	2,279.98	6,554.95		35,054.73	35,054.73
F.3	Sewer and Sanitary Works	1.00	I.s.	21,080.00	5,817.98	581.80	27,479.78	4,121.97	2,198.38	6,320.35		33,800.13	33,800.13
F.4	Pool Plumbing	1.00	I.s.	92,660.00	5,817.98	581.80	99,059.78	14,858.97	7,924.78	22,783.75		121,843.53	121,843.53
F.5	Rainwater Storm	1.00	I.s.	57,340.00	5,817.98	581.80	63,739.78	9,560.97	5,099.18	14,660.15		78,399.93	78,399.93
F.6	Water Storage	1.00	I.s.	65,000.00	5,817.98	581.80	71,399.78	10,709.97	5,711.98	16,421.95		87,821.73	87,821.73
TOTAL OF PART F				90,880.00	17,453.94	1,745.39	110,079.33	16,511.90	8,806.35	25,318.25	-		423,462.76
PART G	ELECTRICAL WORKS												
G.1	Service Entrance, Panel, & Feeders	1.00	I.s.	39,300.00	2,336.64	233.66	41,870.30	6,280.55	3,349.62	9,630.17		51,500.47	51,500.47
G.2	Branch wiring & Conduits	1.00	I.s.	45,030.00	2,336.64	233.66	47,600.30	7,140.05	3,808.02	10,948.07		58,548.37	58,548.37
G.3	Lighting Fixtures and Switches	1.00	I.s.	26,500.00	2,336.64	233.66	29,070.30	4,360.55	2,325.62	6,686.17		35,756.47	35,756.47
G.4	Power - Convenience Outlets	1.00	I.s.	6,110.00	2,336.64	233.66	8,680.30	1,302.05	694.42	1,996.47		10,676.77	10,676.77
G.5	Grounding and Bonding	1.00	I.s.	4,640.00	2,336.64	233.66	7,210.30	1,081.55	576.82	1,658.37		8,868.67	8,868.67
G.6	Pool Electrical	1.00	I.s.	15,200.00	2,336.64	233.66	17,770.30	2,665.55	1,421.62	4,087.17		21,857.47	21,857.47
G.7	Boxes, Fittings and permits	1.00	I.s.	10,320.00	2,336.64	233.66	12,890.30	1,933.55	1,031.22	2,964.77		15,855.07	15,855.07
TOTAL OF PART G				147,100.00	16,356.48	1,635.65	165,092.13	24,763.82	13,207.37	37,971.19	-		203,063.32
Total Project Cost				4,654,690.86	865,399.28	585,104.92	6,105,195.05	905,823.01	483,985.60	1,389,808.61			7,586,231.52

Prepared by:

ENGR. MA. JESSA P. DOLLADO, RMP.

Detailed Cost Estimate

C.6	Gravel Bedding	15.21 cu.m.
-----	----------------	-------------

D.1	Concrete Works: Structural Concrete, Class AA , 4000 PSI (28 Days)	149.50 cu.m.
-----	--	--------------

D.2	Reinforcing Steel (Deformed), Grade 60 (Fy = 414 MPa), for main bars of columns, tie beams, beams, lintel beams, slab, PC1, and footings (PCC)	10,977.41 kg
-----	--	--------------

D.3	Formworks and Falseworks	183.34 sq.m.
-----	--------------------------	--------------

◀ ▶ SOW bom-boq estimate **FINAL BACKUP (rect)** DUPA (+)

Sample Back-Up Computation/ Quantity Take-Off

DETAILED UNIT PRICE ANALYSIS (DUPA)					
Item No. / Description :		D.1	Concrete Works: Structural Concrete, Class		
Unit of measurement :		cu.m.	AA , 4000 PSI (28 Days)		
Output per hour - As Submitted :		3.72			
Output per hour - As evaluated :					
A.1	Designation	No. of Person/	No. of Hour/s	Hourly Rate	Amount (PhP)
	Labor				
	a. Construction Foreman	1	1.00	127.78	127.78
	Skilled Laborer	2	1.00	92.75	185.51
	c Unskilled Laborer	6	1.00	71.54	429.26
	Sub - Total for A.1 - As Submitted				742.55
B.1	Name and Capacity	No. of Units	No. of Hour/s	Hourly Rate	Amount (PhP)
	Equipment				
	a. Concrete Vibrator	2	1	730	1460
	b. Concrete Pump, 100ft3/min	1	0.27	2,789.31	753.1136784
	Minor Tool (10% Labor)				74.26
	Sub - Total for B.1 - As Submitted				2,287.37
C.1	Total (A.1 + B.1) - As Submitted				3,029.92
C.2	Total (A.2 + B.2) - As Evaluated				
D.1	Output per hour - As Submitted	3.72	cu.m.		
D.2	Output per hour - As Evaluated				
E.1	Direct Unit Cost (C.1 / D.1) - As Submitted				814.50
E.2	Direct Unit Cost (C.2 / D.2) - As Evaluated				
F.1	Name and Specification	Unit	Quantity	Unit Cost	Amount (PhP)
	Materials				
	a. Ready Mix Concrete 4000 PSI @ 28 days	cu.m.	1.05	7,500.00	7,875.00
	Sub - Total for F.1 - As Submitted				7,875.00
G.1	Direct Unit Cost (E.1 + F.1)				8,689.50
H.1	Overhead, Contingencies & Miscellaneous (OCM)		15%	of G.1	1,303.42
I.1	Contractors Profit (CP)		8%	of G.1	695.16
K.1	Total Unit Cost		(G.1 + H.1 + I.1 + J.1)		10,688.08

Appendix — Full Deliverables Package

The complete set of project documents is organized as follows and is available on request:

- 01 — Full Architectural, Structural, Electrical, and Plumbing Drawing Set (16+ sheets)
- 02 — Bluebeam Markup Summary (fixtures, quantities, and QC notes by layer)
- 03 — Master Estimating Workbook (SOW, full BOQ, full DUPA, ABC/cost summary, POW, material rates and rentals, manpower and equipment)
- 04 — Design & Estimate QC Addendum (W5 window and railings review)

Supporting Documents Available Upon Request